

Program: Diploma in Renewable Energy	
Course Code: 3291	Course Title: Analog & Optoelectronic Circuits
Semester :3	Credits: 3
Course Category:Programme Core	
Periods per week: 4 (L:3 T:1 P:0)	Periods per semester: 60

Course Objectives:

- To understand the basics of biasing and know different amplifiers.
- To know the basics and types of Oscillators and Timers.
- To introduce the fundamental concept of Operational amplifiers
- To impart basic knowledge of Optoelectronic devices.

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Basic knowledge of Physics		Applied Physics II	2
Basic knowledge of fundamentals of electronics		Fundamentals of Electrical & Electronics Engineering	2

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Classify Optoelectronic Devices	14	Understanding
CO2	Explain the operations of different power amplifiers	15	Understanding
CO3	Classify various Oscillator Circuits.	14	Understanding
CO4	Explain the basic concepts of operational amplifier and its applications	15	Understanding

	Series Test	2	
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CO-PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	2						
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcome	Description	Duration (Hours)	Cognitive Level
CO1	Classify Optoelectronic Devices.		
M1.01	Summarize the basic concepts of semiconductor physics	2	Understanding
M1.02	Outline the operation of PN junction diodes.	4	Understanding
M1.03	Explain the working of optoelectronic devices.	5	Understanding
M1.04	Illustrate the principle of operation of solar cell	3	Understanding
Contents: Semiconductor physics - Semiconductor types- majority and minority carriers -PN junction- formation of PN junction-depletion layer-Energy band diagram PN junction diode- V/I characteristics-Diode equation-Diode parameters -Equivalent circuit of a PN junction Optoelectronic devices: Basic working principle-materials for optoelectronics - Electron emission -Symbols, working and applications of LED, LDR, LCD, photodiode (PIN, APD), photoconductive cell, optocoupler/ optoisolator (phototransistor, photodarlington), Laser diode Solar Cells: Photovoltaic effect -Principle of operation of solar cell - Equivalent Circuit of a Solar Cell -V/I Characteristics			

CO2	Explain the operations of different power amplifiers		
M2.01	Explain the basic operations of a transistor.	4	Understanding
M2.02	Classify the amplifiers based on coupling.	2	Understanding
M2.03	Compare different classes of power amplifiers.	5	Understanding
M2.04	Illustrate the operation of different push-pull amplifiers.	4	Understanding
	Series Test – I	1	
Contents Biasing: Principle of transistor as switch and amplifier, need for biasing, operating point, DC load line, stability factor, voltage divider bias for BJT. Amplifier coupling: coupling schemes- importance of coupling- types - direct, RC coupled and Transformer coupled Power amplifiers: introduction-comparison between voltage amplifier and power amplifier - AC load line-classes of operation - Class A, Class B and Class C (basic operation with diagram)- comparison of various classes Push pull amplifiers -Basic concept of push-pull amplifiers with circuit diagram - waveforms of class B operation-Complementary symmetry Push pull amplifier (basic concept and diagram)			
CO3	Classify various Oscillator Circuits.		
M3.01	Illustrate the concepts of feedback and explain Barkhausen's criterion.	2	Understanding
M3.02	Explain the classification and operation of sinusoidal oscillators.	5	Understanding
M3.03	Explain the classification and operation of non-sinusoidal oscillators.	4	Understanding
M3.04	Summarize the applications of 555 Timer IC.	3	Understanding

Contents:

Feedback- positive and negative feedback(block diagram approach only)

Oscillators - elements in an oscillator circuit-comparison between amplifier and oscillator-Barkhausen's criterion for oscillations-classification of oscillators

Sinusoidal oscillators-classification –Tuned collector Oscillator - Hartley Oscillator – Colpitt's Oscillator-RC-phase shift oscillator - crystal oscillator(basic operation with diagram)

Nonsinusoidal oscillators- classification-multivibrators- astable - bistable- monostable (basic operation with diagram) -applications of oscillators.

Timer ICs– block diagram, pin diagram of 555 timer IC, duty cycle, time constant, applications-Basic idea and functional block diagrams of multivibrators-astable and monostable

CO4	Explain the basic concepts of operational amplifier and its applications		
M 4.01	Explain the parameters and properties of ideal and practical op-amps	3	Understanding
M 4.02	Illustrate the inverting and non-inverting operation of op-amps	4	Understanding
M 4.03	Summarize the applications of operational amplifiers.	4	Understanding
M 4.04	Outline the operation of op-amp comparators and precision rectifiers	4	Understanding
	Series Test – 2	1	

Contents :

Operational amplifiers-Introduction (block diagram approach)- characteristics of ideal and practical op-amps - concept of virtual ground – parameters of op-amp (listing and definitions)

Basic Op-amp circuits -inverting amplifier(derivation of expression for output voltage needed), non-inverting amplifier (derivation not needed)

Op-amp amplifiers - Summing amplifier-subtractor- integrators, differentiators (derivation of output equation needed)

Applications of op-amps-Comparators- positive and negative voltage level detector -

zero crossing detector-schmitt trigger-rectifier using Op-amps- half-wave precision rectifier (operation with circuit and waveforms)

Text / Reference

T/R	Book Title/Author
T1	V.K.Mehta, Principles of Electronics. S Chand & Co.
T2	Ramakant A. Gayakwad, Op-amps and Linear Integrated Circuits, Pearson Education
T3	Optoelectronic devices and systems by M C Gupta
R1	Electronics Analog and Digital by I. J. Nagrath, PHI Learning Pvt. Ltd., 2013 Edition
R2	R.S. Sedha, Applied Electronics. S Chand & Co
R3	B.L.Theraja, Electrical Technology. Vol-IV, S Chand & Co.

Online Resources

Sl.No	Website Link
1	NPTEL/SWAYAM: Analog Electronics Circuits https://nptel.ac.in/courses/117/103/117103063 https://nptel.ac.in/courses/108/105/108105158 https://nptel.ac.in/courses/108/102/108102112
2	Virtual labs: Basic Electronics http://vlabs.iitkgp.ac.in/psac/newlabs2020/vlabiitkgpAE/index.html http://vlabs.iitb.ac.in
3	https://www.electronicshub.org/power-amplifier
4	https://www.elprocus.com